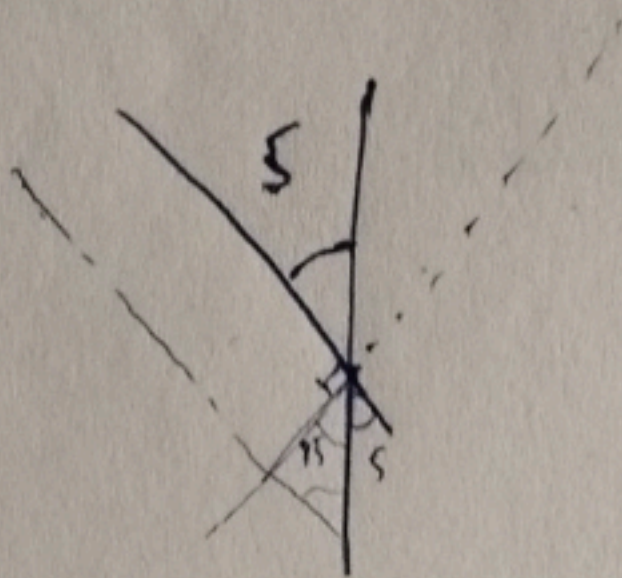
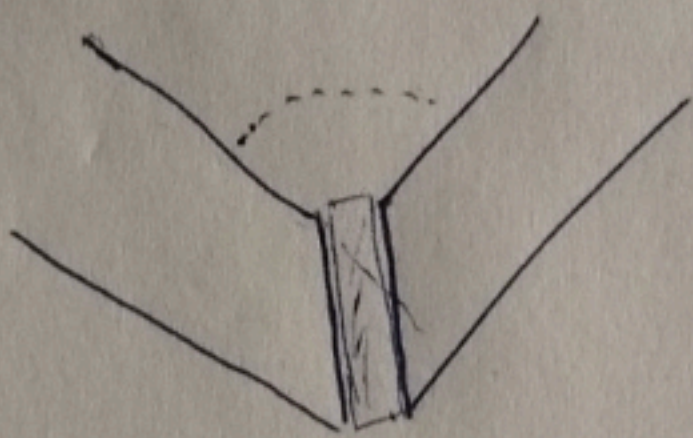




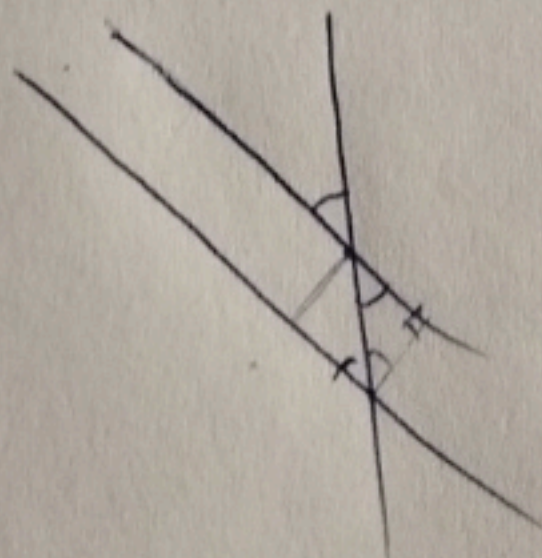
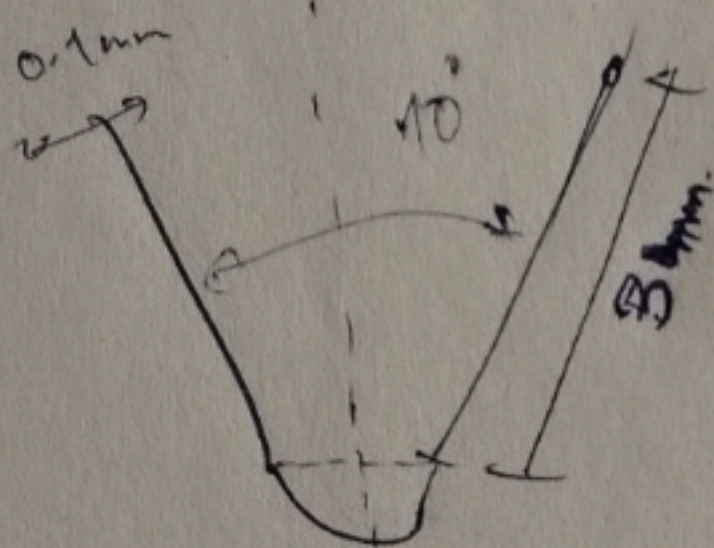
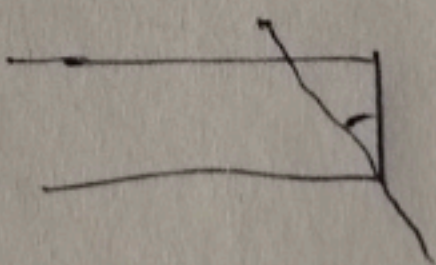
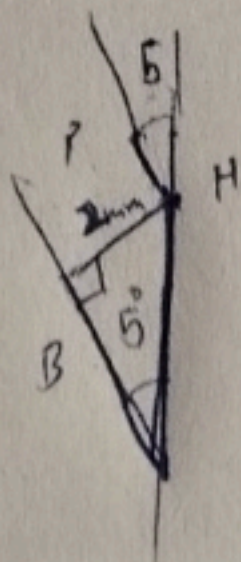
$$\tan = P$$

$$\cos = \frac{B}{H}$$

$$H = \frac{B}{\sin \theta}$$

$$\sin \theta = \frac{P}{H}$$

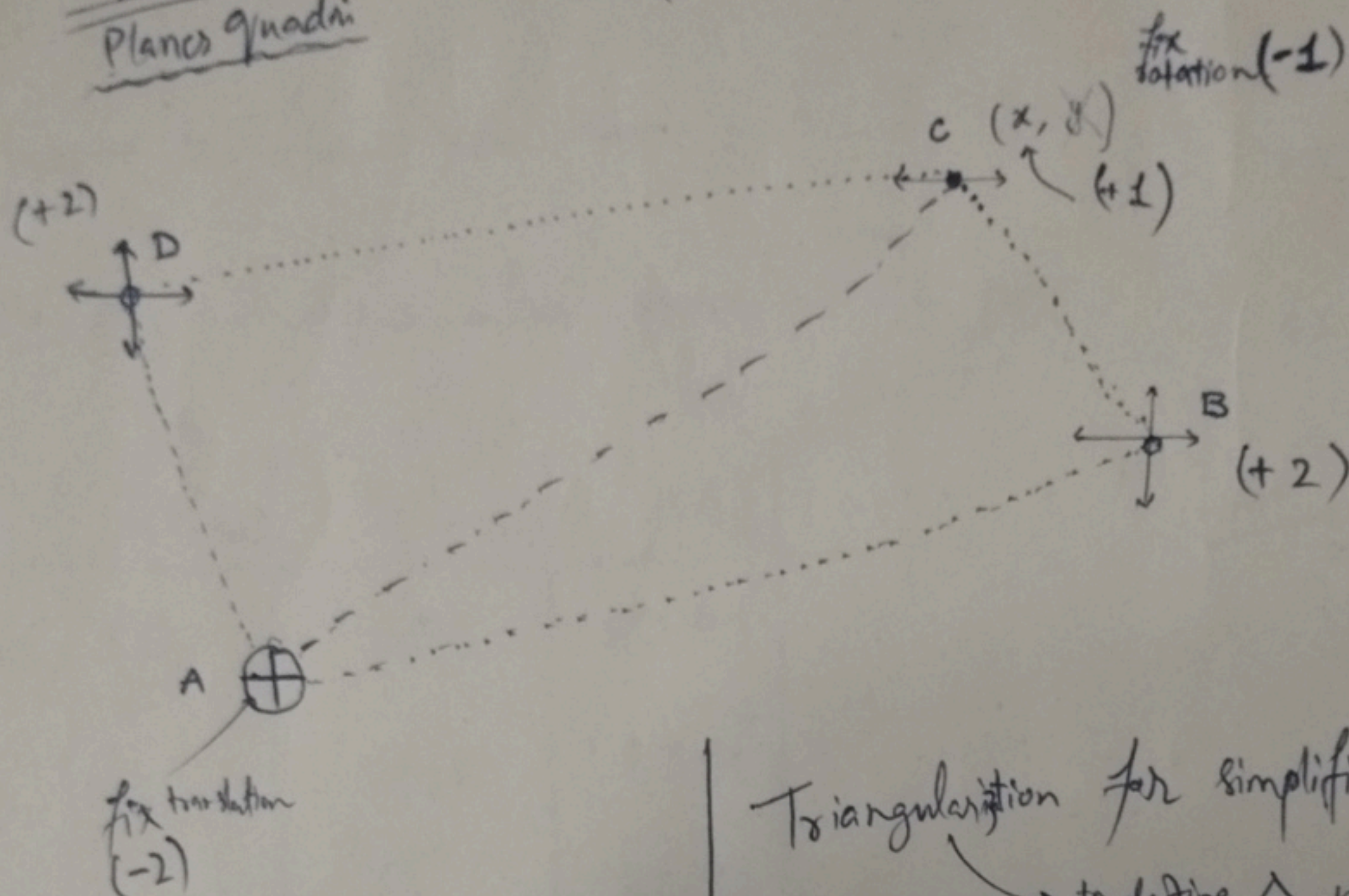
$$H = \frac{P}{\sin \theta} = \frac{2}{\sin 5^\circ}$$



Seminar presentation

— Prof. Ankit + George.

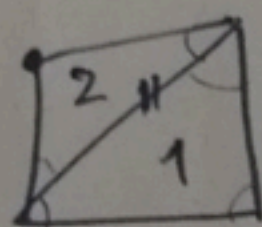
2D
Planar quadrilateral
 Total freedom: $2 \times 4 = 8$ DOF $\xrightarrow{-3} 5$ DOF.
 (or 4 sides & 4 angles)



Min indep.
 DOF
 for quadrilateral
 5
 ↓
 Can remove
 Scaling Parameter
 (only control shape)
 gives 4 parameters

Triangulation for simplification

to define Δ , we need 3 info Max.
 SSA, SSS, SAA



$\Delta_1 = 3$ parameter.

for Δ_2 one side is ^{needed to be} same, so we left with only SS, AA or SA.
 i.e., 2 parameters only.

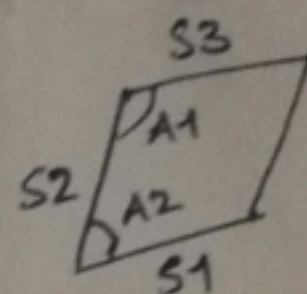
$\therefore$ Total independent parameters = $3 + 2 = 5$

Valid Combination of Parameters:

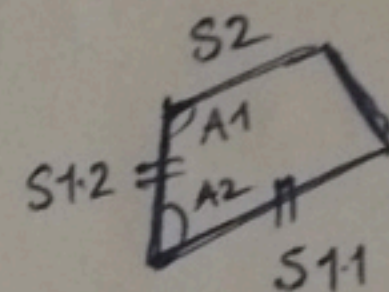
- Max 4 sides
- ① 4 sides + 1 Angle
 - ② 3 sides + 2 Angle
 - ③ 2 ^{Adjacent} sides + 3 Angle.
 (to reduce Redundancy)

4th Angle dictated
 by other three
 as $\angle$ of all inside is 360°

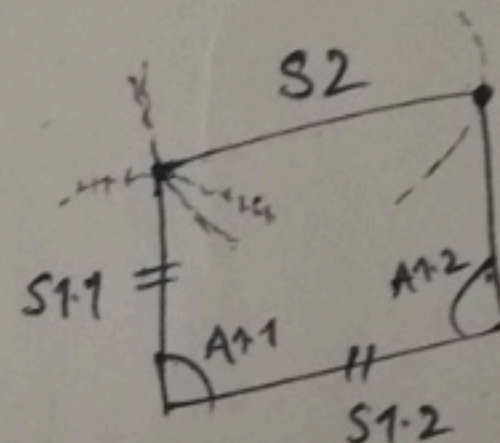
With 3 sides & 2 Angles = 5 DOF



Keeping 2 Adjacent side same = 4 DOF



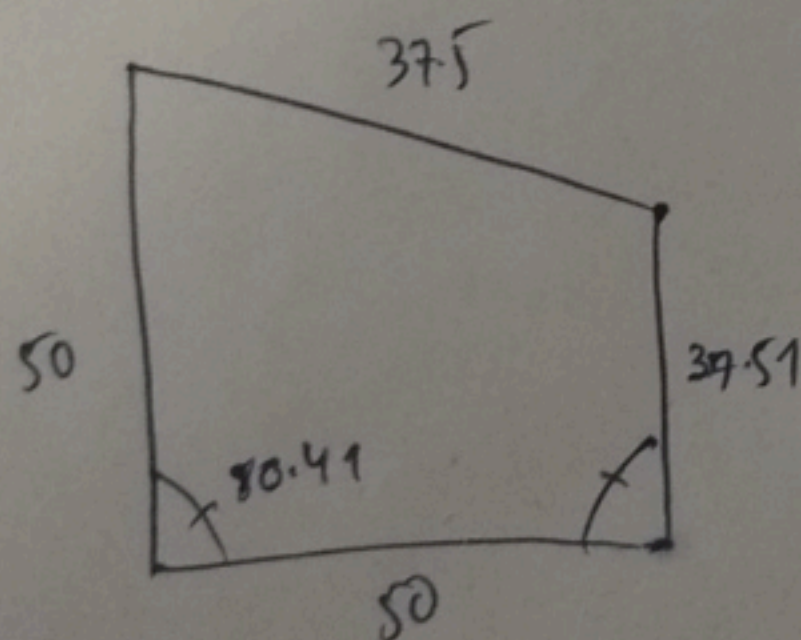
Keeping 2 Angles also same = 3 DOF

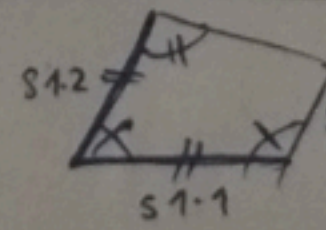


	side S1 (mm)	S2 (mm)	Angle A1 (deg)
Spiral 1	50	37.5	80.41°
" 2	"	41.25	83.91°
" 3	"	45	86.82°

$$\frac{37.51212321}{50} = 0.7502424642 \quad \text{left after. reducing.} \quad 75.02\%$$

$$= 75.02424642\%$$



My CAD ratio : Keeping 3 Angles  & 2 sides (both same)

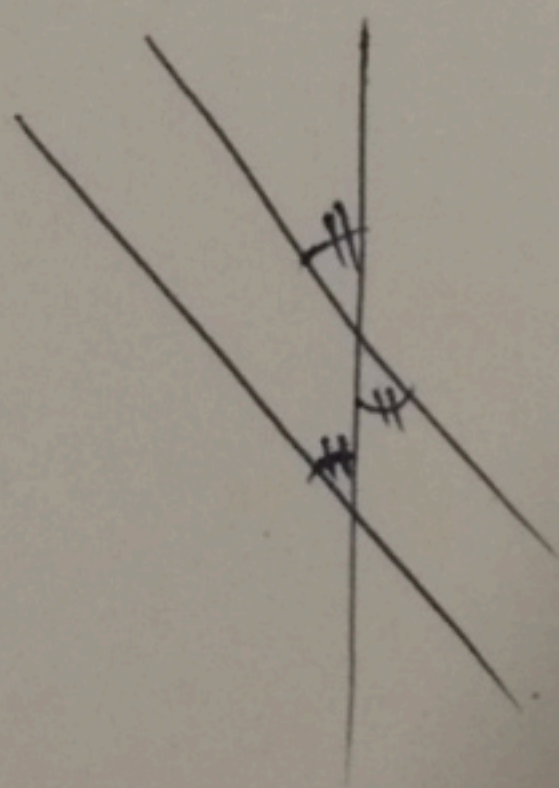
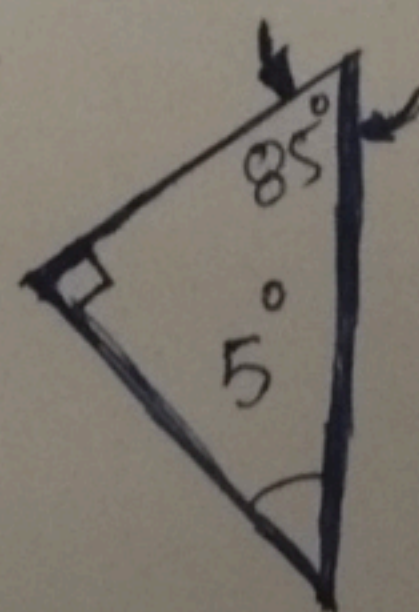
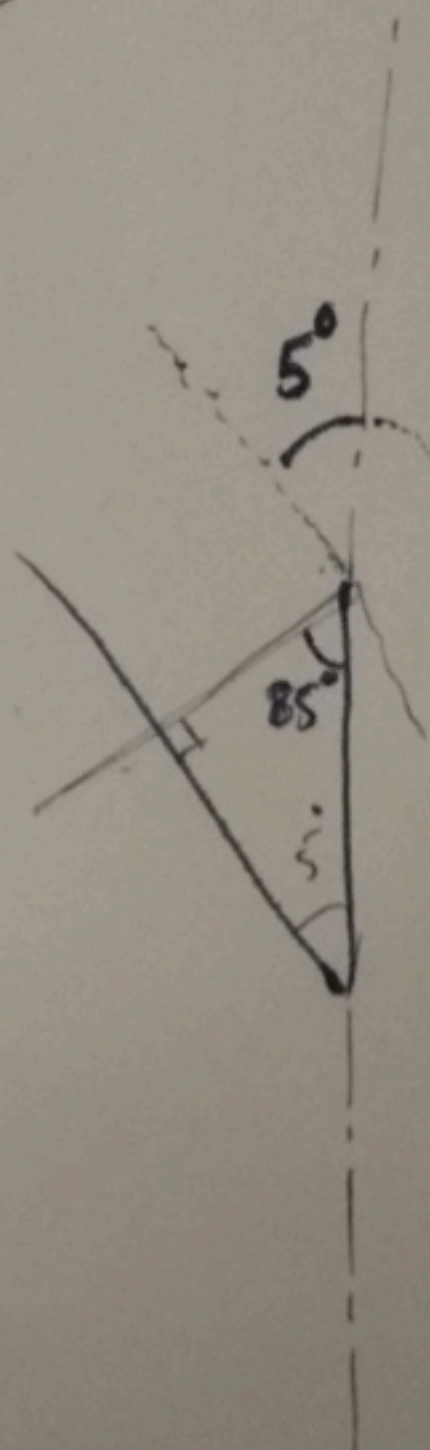
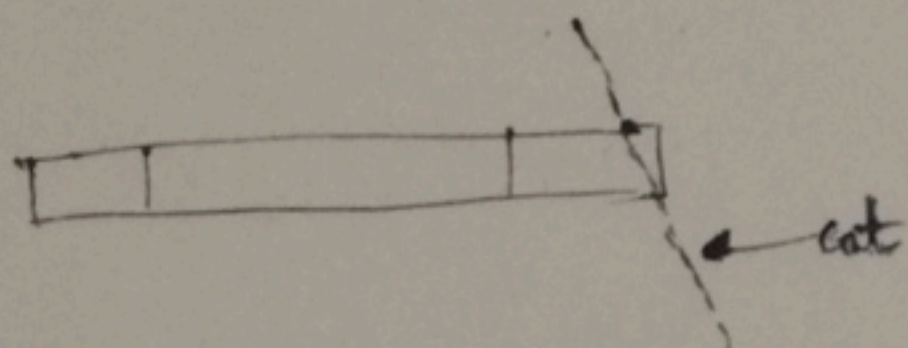
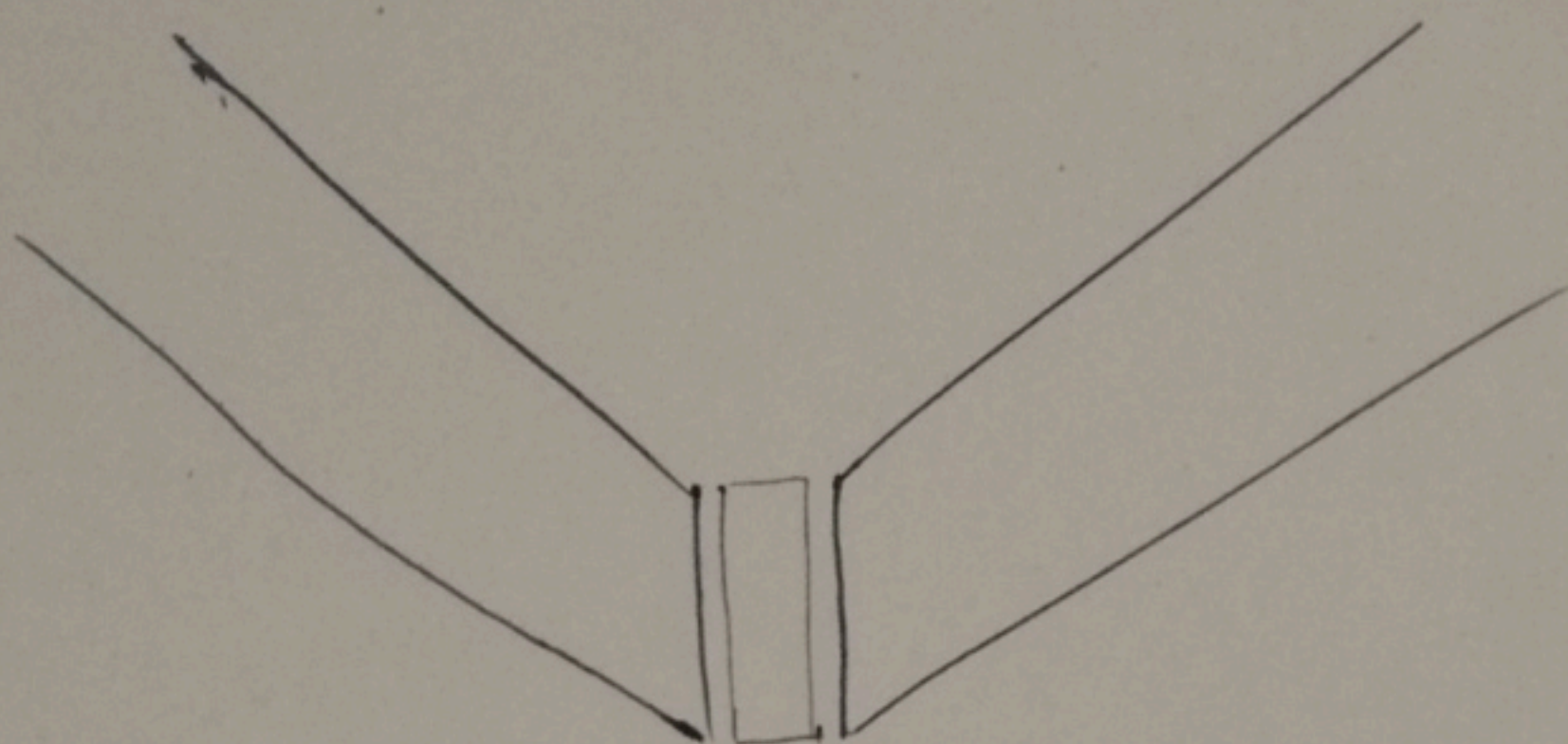
Keeping S1.1 & 1.2 as S3 in consecutive model.

$$50 \rightarrow 37.51212321 \rightarrow 28.14318771 \rightarrow 21.11421446 \rightarrow 15.84628026$$

$$\downarrow$$

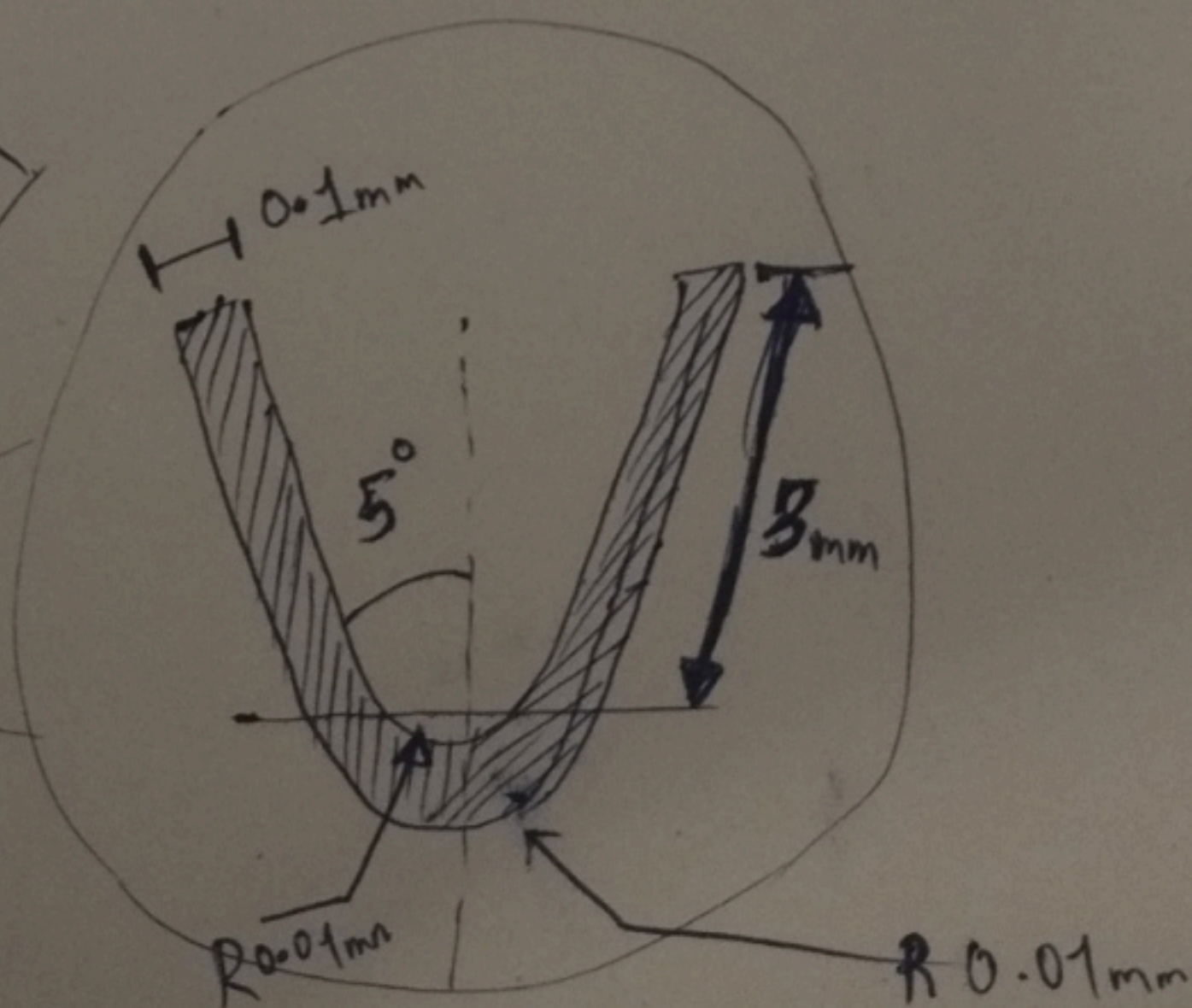
$$11.884426$$

$$\downarrow$$



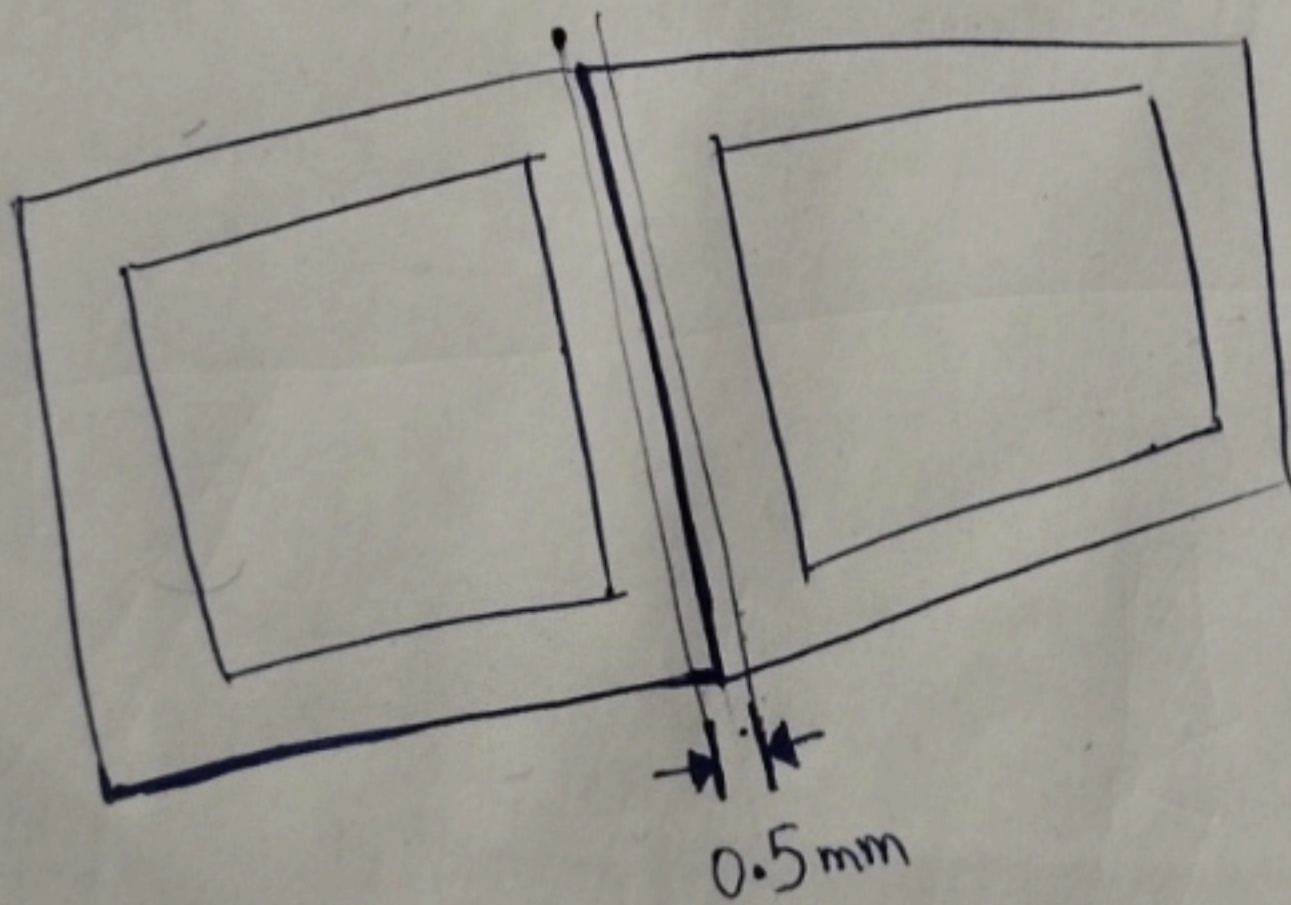
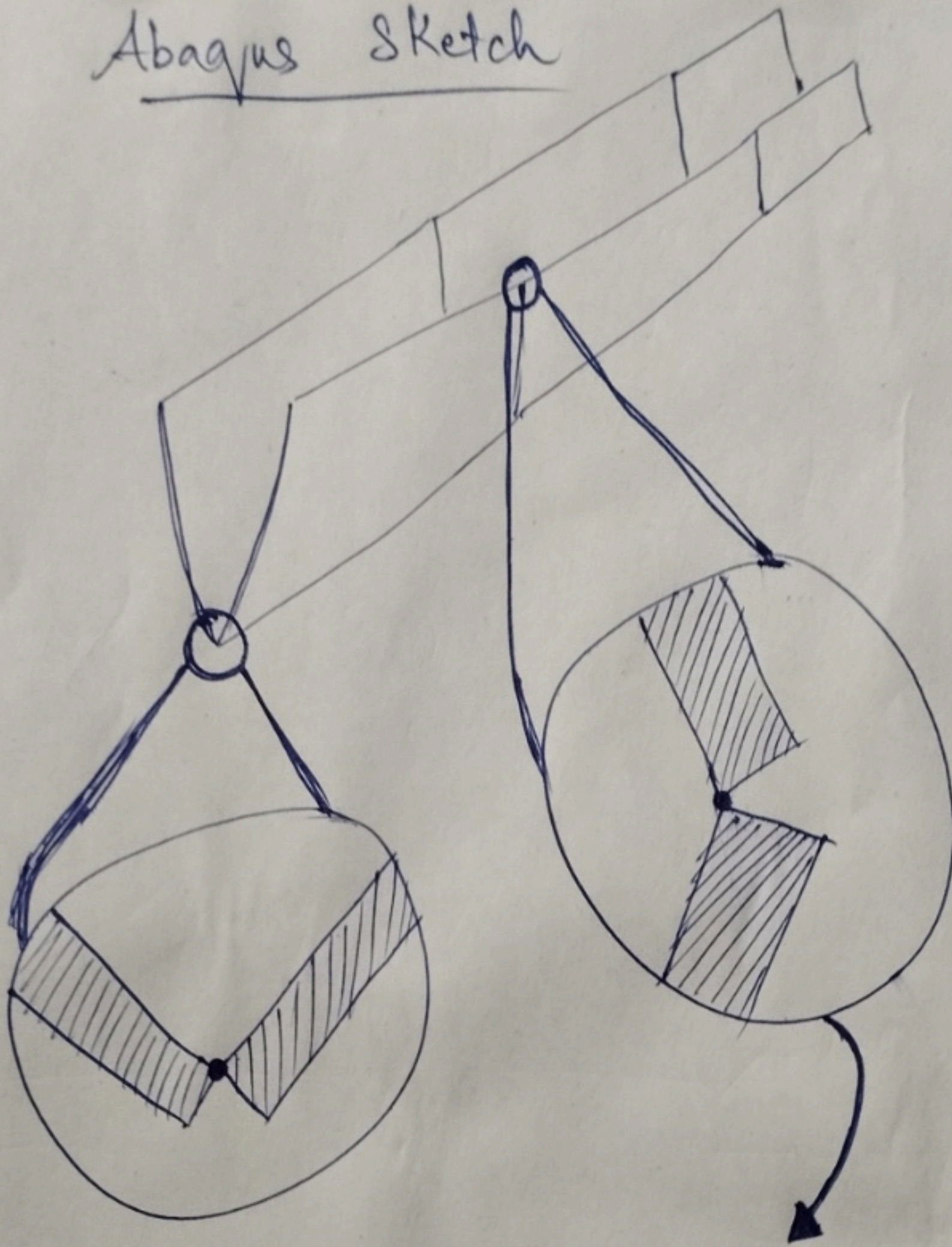
Final
→

Material
parameter
= 0.1 mm thickness



Abaqus Sketch

07 / Feb / 2026



Note: Abaqus.cae file made in Learning Version Won't Open in Abaqus Commercial.

↳ way around it: make Macro (— .py) file and then run it as script in commercial Abaqus.

P.S. don't save file in the Macro.

Properties

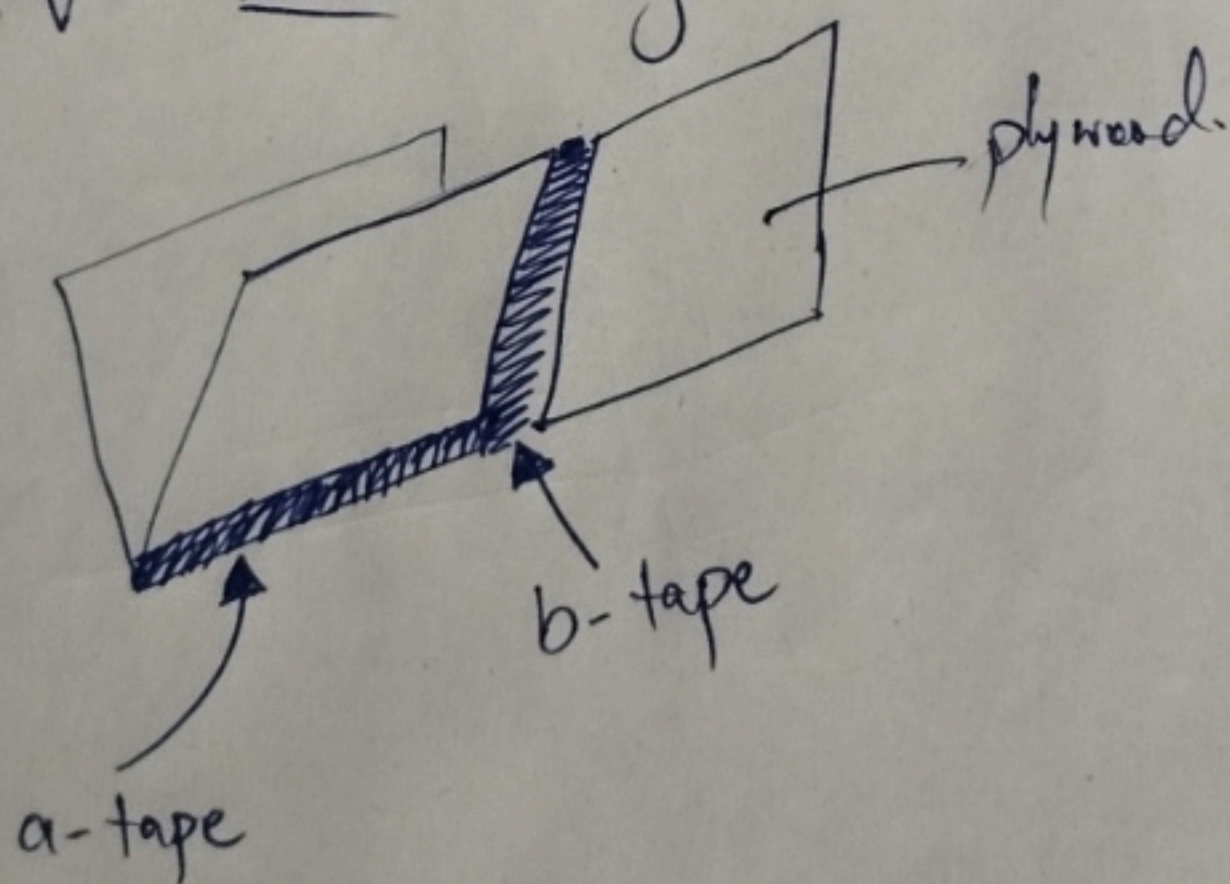
$$E = \frac{\sigma}{\epsilon}$$

- ↳ Masking ~~Kraft Paper~~ tape 10^9
- ↳ $0.5 \text{ GPa} = E$
- ↳ $\nu = 0.34$
- ↳ $\rho = 880 \text{ kg/m}^3$
- ↳ Baltic Birch (Plywood)
 - ↳ $E = 7 \text{ GPa}$
 - ↳ $\nu = 0.13$
 - ↳ $\rho = 680 \text{ kg/m}^3$

3M Technical Data Sheet
 ASTM D882 Standard Test Method for Tearing
 "Handbook of Physical
 & Mechanical Testing of
 Paper & Paperboard."
 "Poisson Ratio of Paper" - Tappi
 Journal
 Bevm, GA et al...

J.K. Chudoba et al
 "Elastic Properties of birch
 plywood". Material &
 Design Vol. 25, 2012

Abaqus set naming.



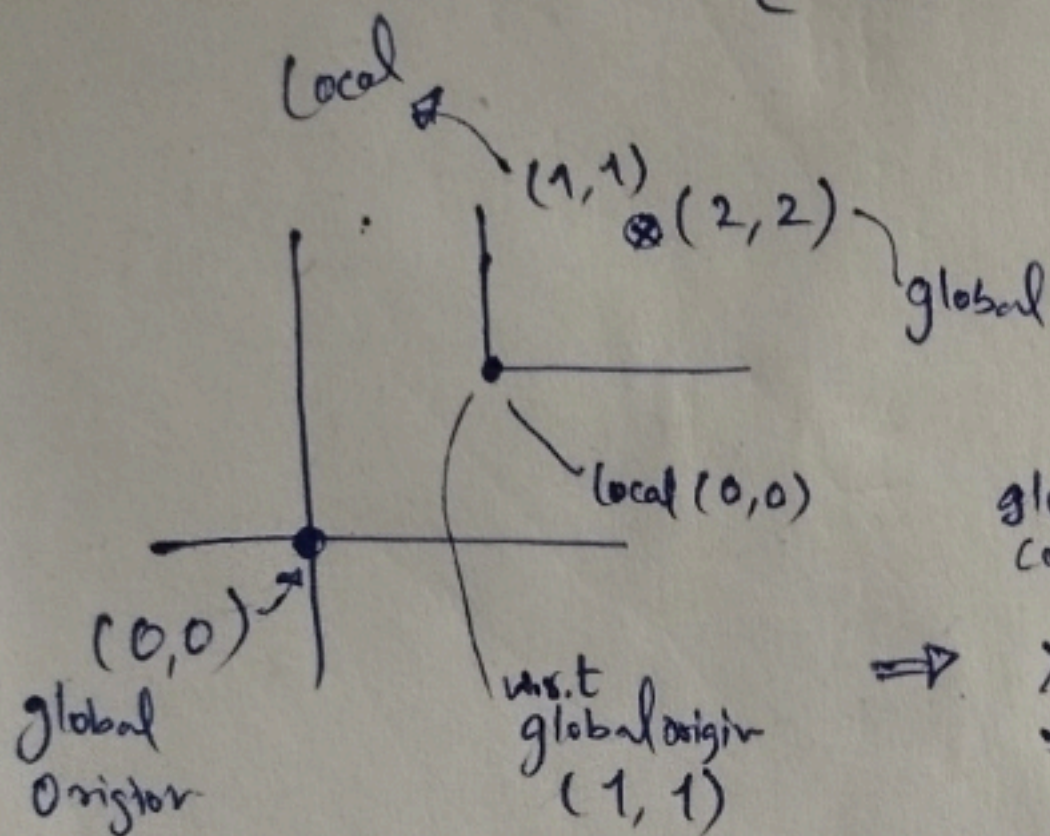
- SI units.
- PPT → Design.
- ↳ Abaqus
 - ↳ Datum
 - ↳ sets → cell
 - ↳ Assign section.
- ↳ Material property.
- ↳ Mesh
 - ↳ Element size: 0.001
 - ↳ Element type: C3D8
 - ↳ Element shape - Hex -
 - ↳ Global seed size: 0.005
 - ↳ Total Element is 20,000
- ↳ Setch of Abaqus
- ↳ video of Abaqus.

Scripting in Abaqus. (Learning)

Node by bounding box

Coordinates:

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} X_0 \\ Y_0 \end{bmatrix}$$



Angle between
global X-axis
& local x-axis
(How much we
need to twist local
axis to align it with global)

Point coordinates
w.r.t local axis

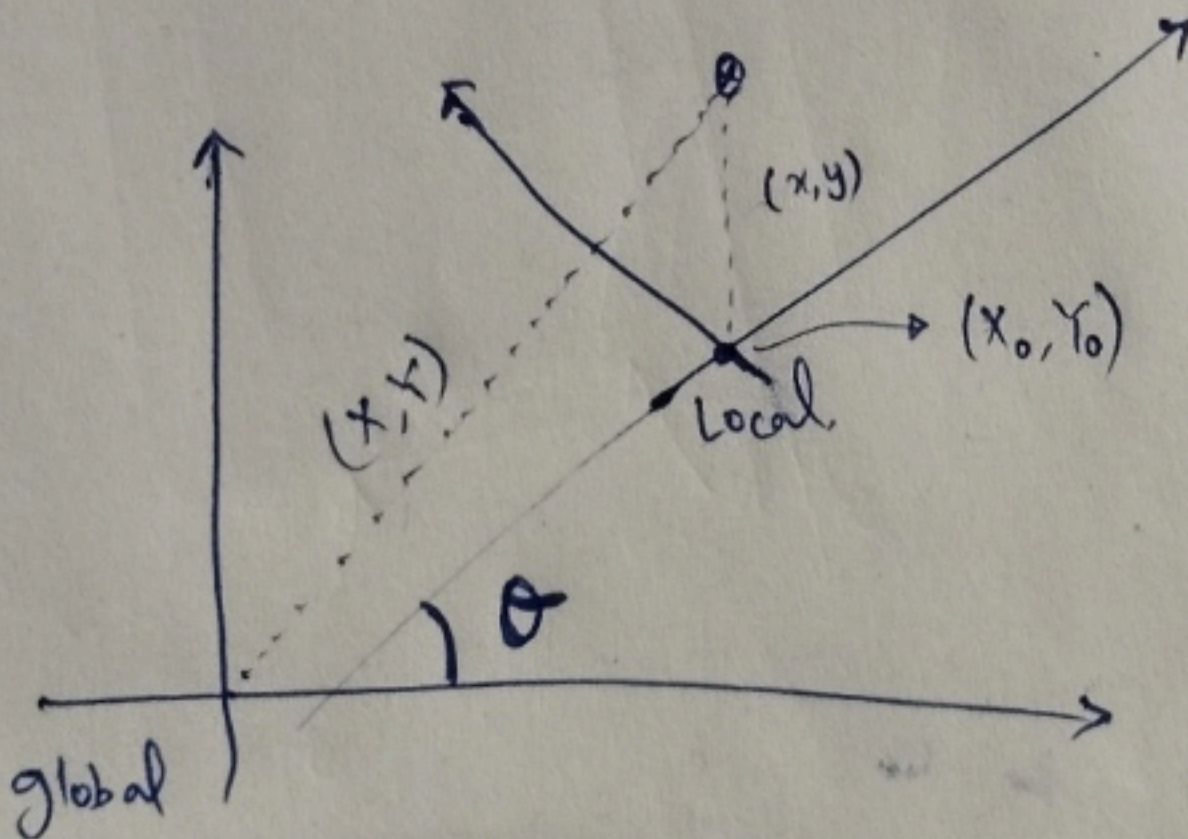
Local Origin
w.r.t global axis.

global coord. = Local Origin w.r.t Origin + local coord.

$$\Rightarrow \begin{aligned} X &= X_0 + x \\ Y &= Y_0 + y \end{aligned}$$

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} X_0 \\ Y_0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} &= 1 + 1 = 2 \\ &= 1 + 1 = 2 \end{aligned}$$



$$X = x \cos\theta - y \sin\theta + X_0$$

$$Y = x \sin\theta + y \cos\theta + Y_0$$

$$\text{if } \theta = 0^\circ \Rightarrow \begin{aligned} \cos\theta &= 1 \\ \sin\theta &= 0 \end{aligned}$$

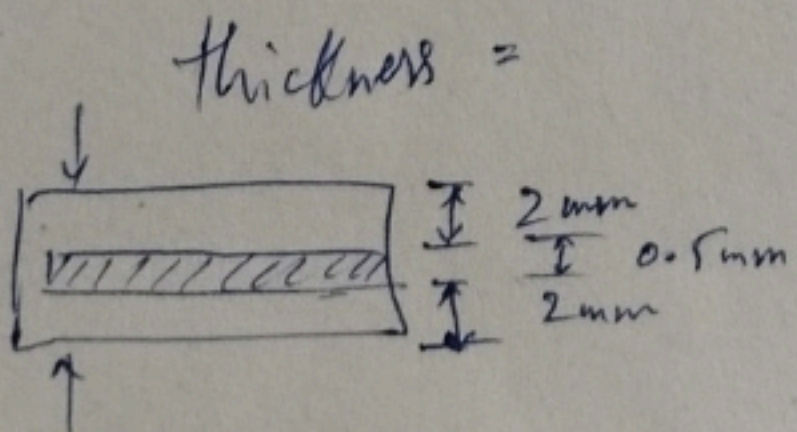
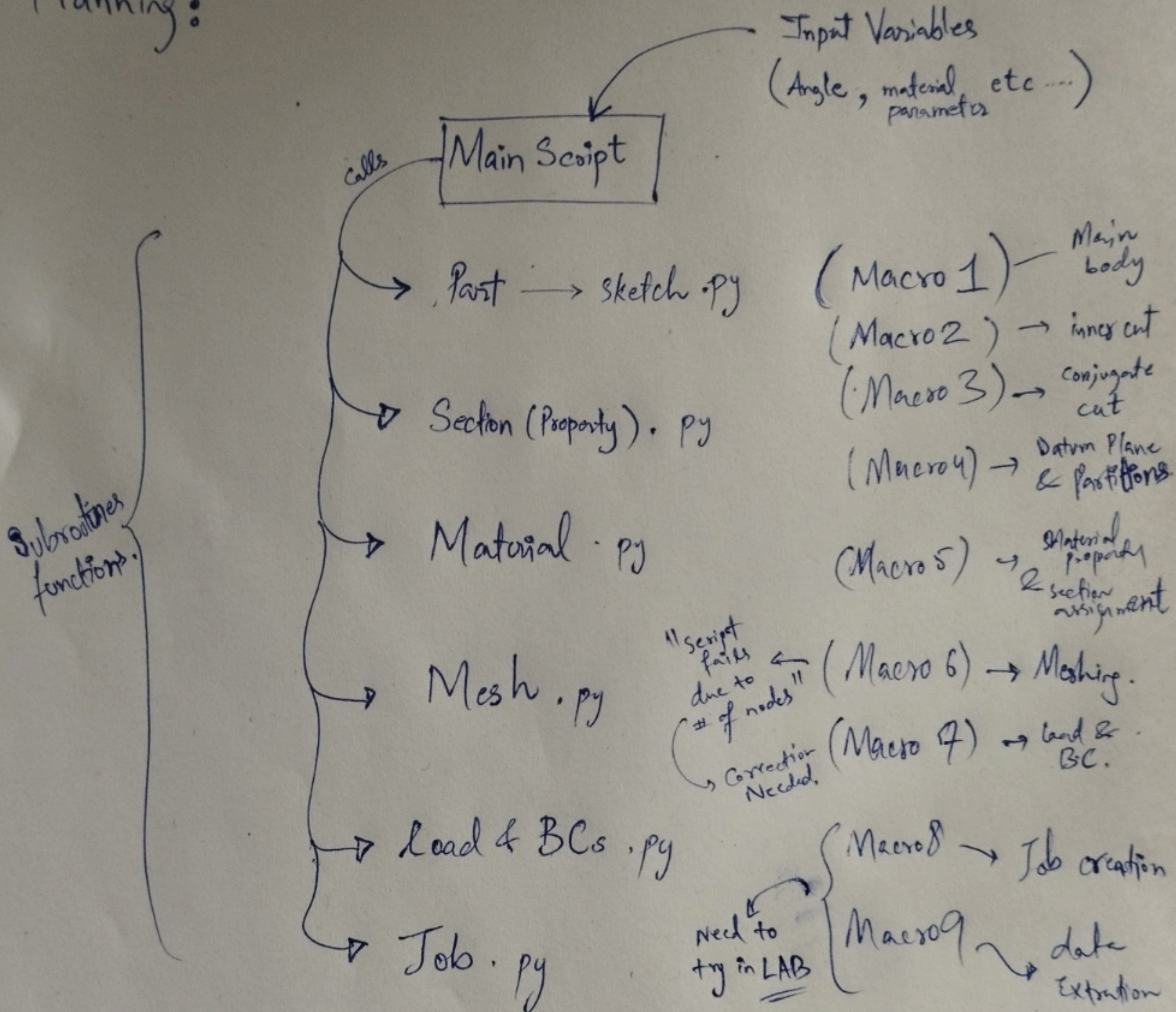
$$\therefore \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\theta = 90^\circ \quad \begin{aligned} \cos 90^\circ &= 0 \\ \sin 90^\circ &= 1 \end{aligned}$$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

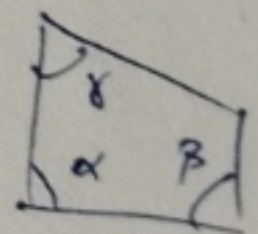
Abaqus Script

Planning:



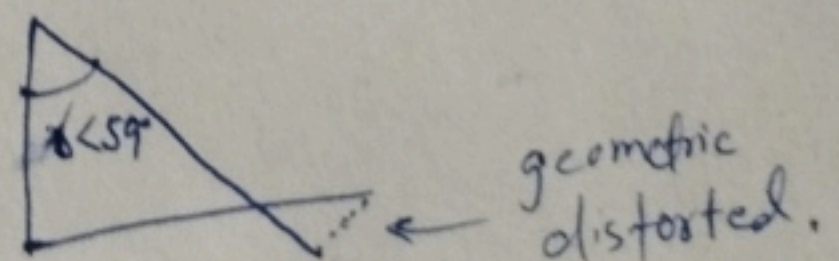
Extrude = 4.5 mm

My finding



This angle "γ"
in script can
decrease to at most $59^\circ \approx 60^\circ$

PS. I am thinking of making it
Not as 3D Extrude but as Shell for better results.



Abaqus Learning

(Error Analysis)

└> Types

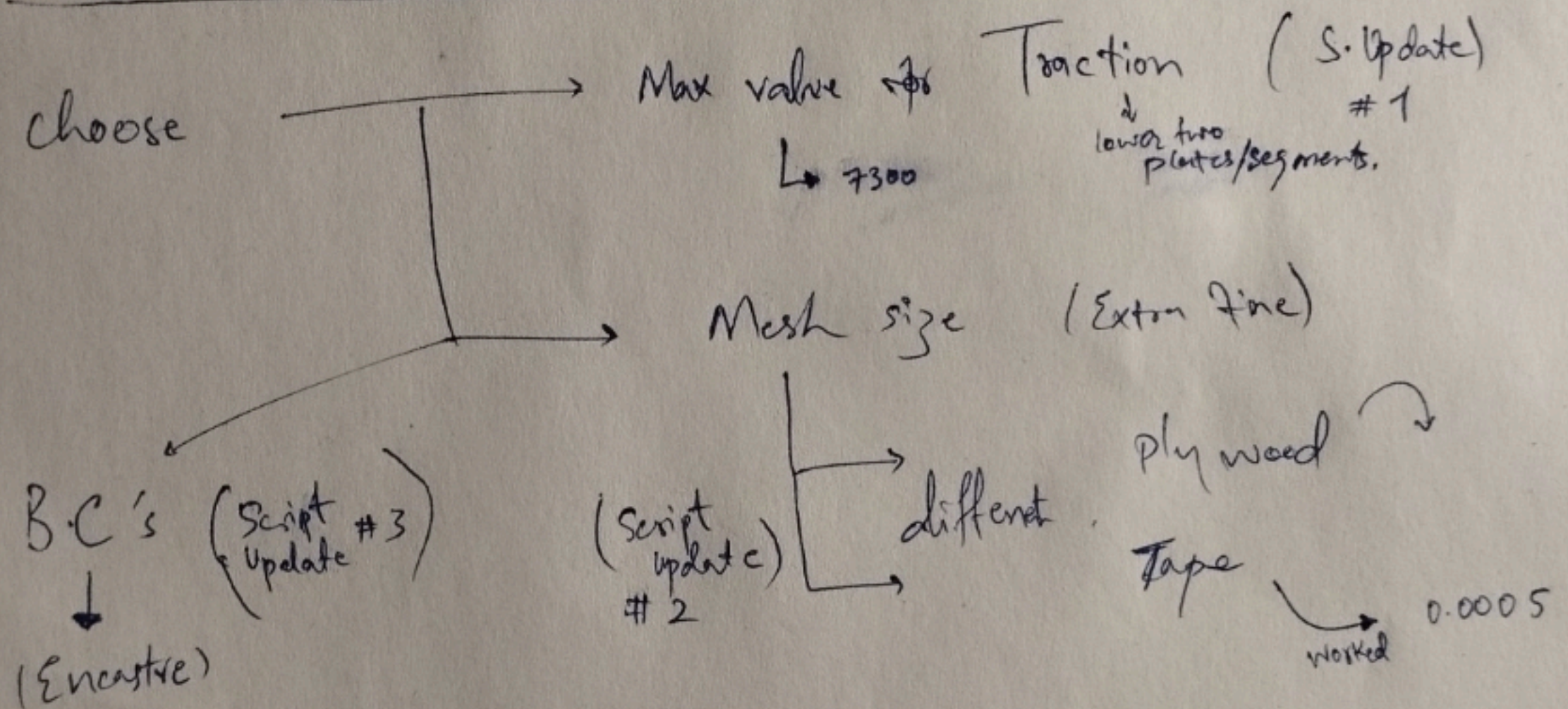
└> Steps / time increments

└> B.C's

└> Monitor jobs

└> Data Extraction

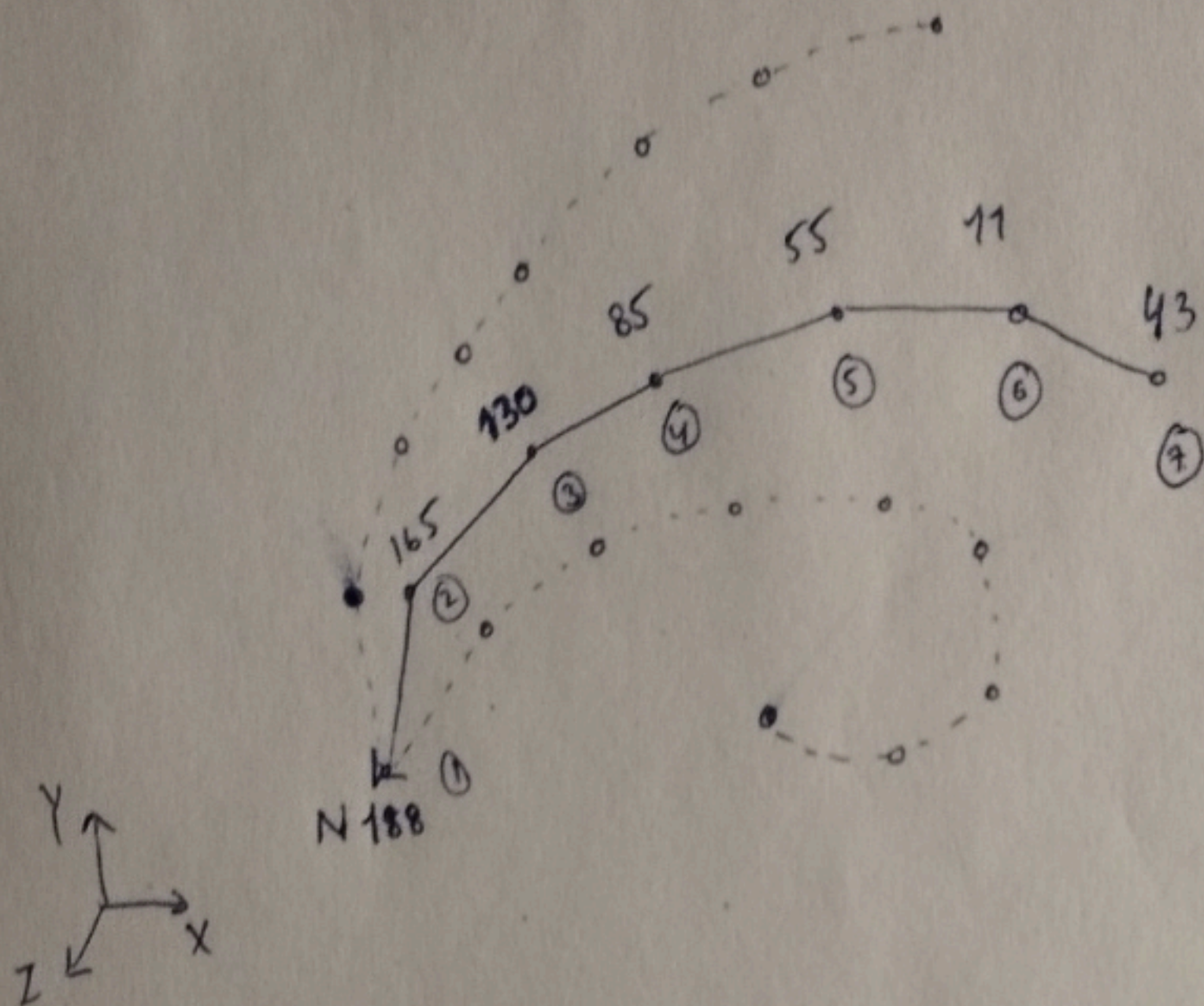
└> Specific Mesh (*) overall global size = 0.0005 (10h sim job)



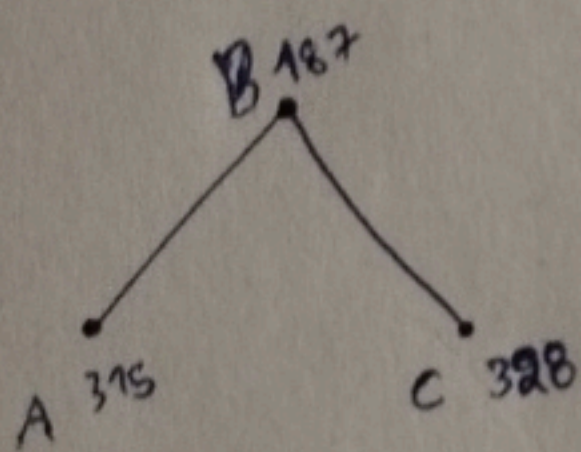
0.0005m
0.5mm

Perfect curl = (out data) — / script - testing — / script-7.cae

└> Job 13: super^{fine} mesh - 8K



* Spiral_Final_CAE $\longrightarrow$ Job 6 $\longrightarrow$ Data for plot.



	Vertex 187	B-187	A315	C-328
x	$1.07417e-4$		$5e-2$	$5e-2$
y	0		0	0
z	$2.25e-3$		$2.5e-3$	$2e-3$